

Embracing The NEW INPUT



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Never has the physical fitness training been so interesting before. From visualising streets of different countries while cycling to adjusting the resistance of the bicycle, virtual reality (VR) is bringing fun and intelligence to physical activities.

Not only the VR technology but also the sensors are acting as miniature input devices to the Internet of Things (IoT) today. One such example is LeChal smart shoes, which use sensor technology to act as a navigator for the visually impaired and trekkers, minimising the use of maps and Internet connection on mobiles.

Call for input technology

From gesture recognition to speech recognition, several new input techniques have come up as strong competitors to the traditional input technology.

Barcode readers, optical mark readers, magnetic ink character recognition readers, optical character recognition readers, magnetic stripe readers, sensors and biometric devices are automatic input devices. Some sensors are passive as these need an external controller to drive them, while some are driven by mechanical principle. Some sensors need digital signal levels, while a few sensors are hybrid in nature, which can be

VR technology for gymming (Image courtesy: <http://ps-media-cache-ak0.pinimg.com>)



used to drive relays or indicators.

As we move forward with the technology, there is always a need to keep improving the design part. For instance, biometric devices have evolved over the years to allow authentication based on facial recognition, voice recognition and vascular recognition. Not just biometrics, there are several other updates in the input technology.

Improved sensor technology is being used at all levels, whether at industrial scale or day-to-day life, saving time by increasing the pace of work.

Project Soli for wearables

T. Anand, co-founder and MD of Knewron Technologies, informs, "The tiny radar chip technology developed by Google in a joint venture is one of the most important upgrades in sensor technology." The chip houses a Soli sensor and an antenna array in an 8x10mm² package, which are managed using the haptic movement of fingers virtually.

Soli sensors can recognise 3D gestures through fingers and hands. To implement these, manufacturers have used a combination of hardware, software and algorithms. Another benefit of Soli sensors is that these hardly require a large bandwidth and high spatial resolution. Thus, the expense of setting up a high-bandwidth system can be overlooked with this 60GHz ISM-band low-power device.

Due to their low-power, low-bandwidth requirements, Soli sensors can be extensively used in wearables, cars, smartphones and IoT devices. Soli's accuracy of tracking and sensing fractional finger movements are helping the developers at Google to build a smart development kit too. So, in the near future, you could be able to start a car just through gesture recognition by a low-power input device.

The product, jointly developed by Google and Infineon, is going to find many applications once it is released. In fact, a joint venture of LG and Google is already working on a wearable based on this technology.